QUANT

Algebra - 1

(Linear, Quadratic Equations & other imp. concepts)

Assignment 02

hitbullseye.com

1. a and b are the roots of $x^2 - x - 3 = 0$. Form the equation whose roots are (3a + 1) and (3b + 1).

1. $x^2 + 5x - 23 = 0$ 2. $x^2 - 5x + 23 = 0$ 3. $x^2 + 5x + 23 = 0$ 4. $x^2 - 5x - 23 = 0$

2. There are some benches in the tenth standard classroom. If 4 students are seated on each bench, 6 students have to stand. If 5 students are seated on each bench, two benches remain vacant. Find the number of benches and the number of students in the class respectively.

1. 18, 60 2. 16, 70 3. 70, 16 4. 12, 70

3. If $\frac{x^2}{yz} + \frac{y^2}{xz} + \frac{z^2}{xy} = 3$, find x + y + z. (Given that $x \neq y \neq z$)

1. 0 2. 1 3. 3 xyz 4. Xyz

4. I have a total of Rs. 300, in 1 rupee, 2 rupee and 5 rupee coins. The number of 2 rupee coins is 3 times the number of 5 rupee coins. The total number of coins is 160. How many 2 Rupee coins do I have?

1. 20 2. 60 3.80 4.40

5. The digits of a two-digit number differ by 3. If the digits are interchanged, and the resulting number is added to the original number, we get 143. What can be the original number?

1. 58 2. 97 3.85 4. Both 1 and 3

6. Meena went to a bank to withdraw Rs. 2000. She asked the cashier to give her Rs. 50 and Rs. 100 notes only. Meena got 25 notes in all. Find how many notes of Rs. 50 and Rs. 100 she received.

1. 10, 15 2.15, 10 3. 5, 20 4. 12, 13

7. The length of a rectangle is less than twice its breadth by 1 cm. The length of its diagonal is 17 cm. Find its length and breadth.

1. 8, 15 2. 6, 8 3. 24, 26 4. 15, 17

8. If 97% of the students are present in a class and 18 students are absent, find the total number of students in the class.

1. 300 2. 1200 3. 600 4. 900

9. Maya, Madhura and Mohsina are friends studying in the same class. In a class test in geography, Maya got 16 out of 25. Madhura got 20. Their average score was 19. How much did Mohsina score?

1. 20 2. 36 3. 75 4. 19

10. In the equation : $2^{x+5} = 2^{x+3} + 6$, the value of x is :

1. 2 2. 4 3. –2 4. 3

Answer key & Explanations

1.	4	Given a + b = $\frac{-(-1)}{1} = 1$ and ab = $\frac{-3}{1} = -3$ Equation whose roots are (3a+1), (3b+1) is x^2 – (sum of roots) x + (product of roots) = 0 i.e. $x^2 - (3a + 3b + 2)x + (9ab + 3a + 3b + 1) = 0$, $\Rightarrow x^2 - 5x - 23 = 0$. Thus 4th option.
2.	2	Go by options. Number of benches is less than number of students. So 3rd option is ruled out. 2nd option is the correct answer because when 4 students are seated on each bench $\therefore$ Total number of students sitting on the benches $= 16 \times 4 = 64$. $\therefore$ 6 students have to stand. Also when 5 students are sitting on each bench, total benches used = $\frac{70}{5}$ = 14. Hence 2 benches remain vacant. $\therefore$ Number of benches = 16 & no. of students = 70.
3.	1	Taking LCM of the LHS you get $x^3 + y^3 + z^3 = 3xyz$. As per the formula of $a^3 + b^3 + c^3 - 3abc$, If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$, as the second part is what you get after LCM, implies $x + y + z = 0$.
4.	2	Let X, Y and Z be the number of Re.1, Rs. 2 and Rs. 5 coins respectively. We can make equations as X + 2Y + 5Z = 300 X + Y + Z = 160 And Y = 3Z ; Solving these we get X = 80, Y = 60, Z = 20 OR On seeing options we can observe that the 1st three options sum up to total of 160 coins. If there are 60 2 rupee coins, it makes a sum of Rs. 120. The remaining sum of Rs. 180 has to be made from Re.1 and Rs. 5 coins. The given number of Rs. 2 coins is 3 times the number of Rs. 5 coins, So Rs. 5 coins would be 20 in number and remaining 80 coins would be of Re.1 denomination.
5.	4	We can simply eliminate option 2 as the digits of the number do not vary by 3. Going by options, both 1 and 3 satisfy the given conditions.
6.	1	If she has 10 Rs. 50 notes and 15 Rs. 100 notes, then the total sum of money will be $10 \times 50 + 100 \times 15$ = Rs. 2000, which is equal to the amount she wants to withdraw. Hence option 1.
7.	1	L = 2B – 1, Diagonal = 17. $D^2 = L^2 + B^2$ $\Rightarrow 289 = (2B - 1)^2 + B^2 \Rightarrow B = 8$ & L = 15. Also 8, 15, 17 is a Pythagorean triplet. So answer can be reached directly.
8.	3	3 % of x = 18. x = 600
9.	4	Let Mohsina score x marks, So $\frac{16 + 20 + x}{75} = 19$
10.	3	$2^{x+5} = 2^{x+3} + 6$, $2^5 2^x = 2^x \times 2^3 + 6$, $2^x(24) = 6$, $2x = \frac{6}{24}$, $2^x = 2^{-2}$ and x = -2.OR Go by options.